

CHENYU ZHANG

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EDUCATION

Massachusetts Institute of Technology Ph.D. Social and Engineering Systems and Statistics Presidential Fellow (Sole recipient in cohort)	Cambridge, MA 2024 - present
Columbia University M.S. Data Science	New York, NY 2022 - 2024
Fudan University B.S. Mathematics and Applied Mathematics Honors Student of Su Buqing Top Talent Program	Shanghai, China 2018 - 2022

PUBLICATIONS

- [1] **Chenyu Zhang**, Rachel Luo, Boyi Li, Anjali Parashar, Marco Pavone, and Apoorva Sharma. Careful Judge: Safe and efficient human-AI collaborative decision making. *Preprint*, 2026.
- [2] **Chenyu Zhang**, Rohit Parashar, and Saurabh Amin. Network intervention by polling strategic agents. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
- [3] **Chenyu Zhang** and Navid Azizan. Personalized collaborative learning with affinity-based variance reduction. *International Conference on Learning Representations (ICLR)*, 2026.
- [4] **Chenyu Zhang** and Rujun Jiang. Riemannian adaptive regularized Newton methods with Hölder continuous Hessians. *Computational Optimization and Applications*, 2025.
- [5] **Chenyu Zhang**, Xu Chen, and Xuan Di. Stochastic semi-gradient descent for learning mean field games with population-aware function approximation. *International Conference on Learning Representations (ICLR)*, 2025.
- [6] **Chenyu Zhang**, Rufeng Xiao, Wen Huang, and Rujun Jiang. Riemannian trust region methods for SC^1 minimization. *Journal of Scientific Computing*, 2024.
- [7] Fuzhong Zhou, **Chenyu Zhang**, Xu Chen, and Xuan Di. Graphon mean field games with a representative player: analysis and learning algorithms. *International Conference on Machine Learning (ICML)*, 2024.
- [8] **Chenyu Zhang**, Han Wang, Aritra Mitra, and James Anderson. Finite-time analysis of on-policy heterogeneous federated reinforcement learning. *International Conference on Learning Representations (ICLR)*, 2024.
- [9] **Chenyu Zhang**, Xu Chen, and Xuan Di. A single online agent can efficiently learn mean field games. *European Conference on Artificial Intelligence (ECAI)*, 2024.

RESEARCH EXPERIENCE

Human-AI Collaborative Decision Making [1] Research intern, supervised by Dr. Apoorva Sharma Co-supervised by Prof. Marco Pavone	May 2026 - September 2026 NVIDIA Research, CA Dept. AA, Stanford University & NVIDIA Research, CA
• Designed an end-to-end human-AI decision-making pipeline prioritizing human query efficiency under user-specified risk constraints, with support for flexible offline-online model lifecycles. • Developed a modular framework for calibrated and adaptive AI improvement from human feedback, with theoretical model-agnostic safety guarantees. • Built and evaluated the system on four safety-critical benchmarks spanning language, vision, driving, and robotics, reducing human queries by 25–81% relative to baseline methods.	

Multi-Activity Network Games [2]

February 2025 - Present

Ph.D. student, advised by Prof. **Saurabh Amin** LIDS, Massachusetts Institute of Technology, MA

- Developed a polling-based method for learning optimal interventions in networks with strategic agents, with improved robustness and sample efficiency.

Personalized Collaborative Learning [3]

September 2024 - Present

Ph.D. student, advised by Prof. **Navid Azizan** LIDS, Massachusetts Institute of Technology, MA

- Designed the first multi-agent learning framework that enables collaboration between arbitrarily heterogeneous agents and yields fully personalized solutions with an adaptive, affinity-based speedup.

Neural Networks

May 2023 - December 2023

Research assistant, advised by Prof. **John Wright** Dept. EE&APAM, Columbia University, NY

- Analyzed limitations of Transformer architectures and designed alternative attention mechanisms using tensor kernels and invariant similarity weights.
- Conducted empirical studies of patch-manifold intrinsic dimension to characterize task-dependent optimal patch sizes for vision models.

Mean Field Games [5,7,9]

May 2023 - August 2024

Research assistant, advised by Prof. **Sharon Di** Dept. CEEM, Columbia University, NY

- Pioneered the first fully online single-agent model-free methods for learning mean field games (MFGs), achieving sample efficiency and stability enhancements without auxiliary mechanisms.
- Extended these methods to linear MFGs with continuous state-action spaces and graphon MFGs with large heterogeneous populations.

Federated Reinforcement Learning [8]

September 2022 - December 2023

Research assistant, advised by Prof. **James Anderson** Dept. EE, Columbia University, NY

Co-advised by Prof. **Aritra Mitra** Dept. ECE, North Carolina State University, NC

- Developed a novel on-policy federated reinforcement learning method and established its finite-time error bounds, demonstrating linear convergence speedup despite the presence of environmental heterogeneity.

Nonsmooth Nonconvex Manifold Optimization [4,6]

October 2021 - September 2022

Research assistant, advised by Prof. **Rujun Jiang** Dept. Data Science, Fudan University, CN

Co-advised by Prof. **Wen Huang** Dept. Mathematics, Xiamen University, CN

- Developed the first Riemannian trust-region method tailored for minimizing nonconvex functions on manifolds with a semismooth gradient field, supported by rigorous global and local convergence analysis.
- Implemented the semismooth Riemannian trust-region method in solving augmented Lagrangian method subproblems on manifolds, demonstrating its superiority through two numerical experiments.
- Extended the methodology to formulate a Riemannian adaptive regularized Newton methods framework and established its sharp worst-case iteration and operation complexities.

Reinforcement Learning with Partial Observability

March 2021 - January 2022

Research assistant, advised by Prof. **Zhaoran Wang** Dept. of IEMS&CS, Northwestern University, IL

Co-advised by Prof. **Zhuoran Yang** Dept. of Stat&Data Science, Yale University, CT

- Devised reinforcement learning exploration methods under partial observability without reward feedback.
- Enhanced the framework with linear function approximation to continuous state and observation spaces.

TEACHING EXPERIENCE

15/6/IDS.C57 Optimization Methods	<i>Fall 2025</i>
<i>Teaching assistant, with Prof. Alexandre Jacquillat and Prof. Saurabh Amin</i>	<i>MIT Sloan, MA</i>
6.431x Probability	<i>Summer 2025</i>
<i>Teaching assistant, with Prof. John Tsitsiklis and Prof. Patrick Jaillet</i>	<i>Dept. EECS, MIT, MA</i>
18.6501x Fundamentals of Statistics	<i>Summer 2025</i>
<i>Teaching assistant, with Prof. Philippe Rigollet</i>	<i>Dept. Mathematics, MIT, MA</i>
COMS 4771 Machine Learning	<i>Fall 2023</i>
<i>Teaching assistant, with Prof. Daniel Hsu</i>	<i>Dept. CS, Columbia University, NY</i>
ORCS 4529 Reinforcement Learning	<i>Fall 2023</i>
<i>Teaching assistant, with Prof. Shipra Agrawal</i>	<i>Dept. IEOR, Columbia University, NY</i>
EEOR 4650 Convex Optimization	<i>Fall 2023</i>
<i>Teaching assistant, with Prof. James Anderson</i>	<i>Dept. EE, Columbia University, NY</i>
CSOR 4231 Analysis of Algorithms	<i>Spring 2023</i>
<i>Teaching assistant, with Prof. Eleni Drinea</i>	<i>Dept. CS, Columbia University, NY</i>

SCHOLARSHIP & AWARD

• MathWorks Engineering Fellowship, MathWorks	<i>2026-2027</i>
• Presidential Fellowship, Massachusetts Institute of Technology	<i>2024</i>
• Honors Student of Top Talent Program, Fudan University	<i>2022</i>
• Undergraduate Merit Scholarship, Fudan University	<i>2018-2019, 2019-2020, 2020-2021, 2021-2022</i>
• Undergraduate Major Scholarship, Fudan University	<i>2019-2020, 2020-2021, 2021-2022</i>
• Freshman Scholarship, Fudan University	<i>2018</i>

INVITED TALKS

• INFORMS Annual Meeting	<i>San Francisco, CA, November 2026</i>
• Learning from Heterogeneous Sources workshop	<i>Simons Institute, CA, February 2026</i>
• Jump AI Research Symposium (Inaugural)	<i>Jump Trading, NY, November 2025</i>

TECHNICAL SKILLS

Languages	Python, JavaScript, C/C++, MATLAB, R, Lua, Shell
Frameworks & Libraries	PyTorch, TensorFlow, NumPy, scikit-learn, pandas, Keras, Flask
Tools	CUDA, Git, Linux, Docker, Google Cloud Platform, Wandb

RELEVANT COURSEWORK

Selected Graduate Courses			
Decision Theory*	A	Network Science*	A+
Game Theory*	A	Algorithms for Data Science	A+
Modern Mathematical Statistics*	A	Modern Control Theory*	A+
Machine Learning	A	Reinforcement Learning*	A+
High-Dimensional Probability*	A	Probability and Statistics for Data Science	A+
Selected Undergraduate Courses			
Numerical Linear Algebra and Optimization [†]	A	Deep Learning	A
Numerical Solution to Differential Equations	A	Methods of Optimization	A
Numerical Analysis	A-	Probability Theory	A
An Introduction to Differential Manifolds [†]	A-	Advanced Algebra	A
Complex Analysis	A	Fundamentals of Mechanics	A

Selected Seminars

Matrix Analysis

Heuristic Optimization Algorithms

Advanced Mathematical Analysis

Global Differential Geometry

Differential Geometry of Curves and Surfaces

Convex Optimization

Complex Analysis

Principles of Mathematical Analysis

Differential Manifolds and Differential Topology

Non-Euclidean Geometry and Point Set Topology

SERVICE

Journal Reviewer

IMA Journal of Numerical Analysis, IEEE Transactions on Signal Processing, IEEE Transactions on Automatic Control, IEEE Transactions on Control of Network Systems

Conference Reviewer

ICLR, ICML, NeurIPS, AISTATS, AAAI, CDC, L4DC

*Doctoral level courses

†Honors courses